

Gas–star segregation and the resilience of King-family clusters to gas expulsion: a semi-analytic study

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ABSTRACT

Context. Star clusters survive instantaneous gas expulsion more easily when star formation proceeds with a centrally peaked star formation efficiency (SFE), because the stars are then born more concentrated than the gas they must unbind from. The observationally motivated, spatially truncated King family has not been mapped in this respect, even though it is the family adopted by upcoming N -body campaigns.

Aims. We quantify how the survivability of an embedded King-model cluster depends on the concentration parameter W_0 , identify the physical mechanism that controls it, and provide calibrated initial-condition predictions for direct N -body simulations.

Methods. Assuming a constant SFE per free-fall time, we derive in closed form (up to quadratures) the residual-gas density profile for King models with $W_0 = 0.5$ –20, compute the post-expulsion virial ratio – validated against the effective SFEs inferred at the published Plummer and Dehnen N -body thresholds – and introduce a scale-free criterion that removes gas whose local free-fall time exceeds the star formation duration. Independently, we compute the bound mass fraction from a frozen stellar distribution function obtained by an analytic Eddington inversion in the total star–gas potential, and cross-check it against Monte Carlo realizations drawn from the same distribution function.

Results. The virial criterion yields an SFE threshold that is a non-monotonic function of W_0 , falling to a minimum of ≈ 0.11 near $W_0 \approx 8.8$ for the reference choice $e\text{SFE}_{\text{surv}} = 1/3$, and rising again toward high concentration. The dependence reduces exactly to a single dimensionless parameter, the normalized star–gas cross energy per unit gas mass, whose minimum is inherited from the maximum of the King structural ratio r_t/r_h , where gas–star segregation is greatest. The result is robust to the free-fall-time truncation. The frozen-DF analysis reproduces the same non-monotonic trend by an independent criterion, and predicts that highly concentrated models keep a self-bound central seed down to global SFEs of a few per cent. The survival value of the effective SFE is shown to be non-universal across density profiles.

Conclusions. Clusters born with intermediate King concentrations ($W_0 \approx 8$ –9) are the most resilient to instantaneous gas expulsion. The predicted ordering with W_0 – rather than any single absolute threshold – is the sharp, falsifiable result that the N -body campaign of Paper I will test.

Key words. galaxies: star clusters: general – methods: analytical – methods: numerical – stars: formation – stars: kinematics and dynamics

1. Introduction

The response of a young star cluster to the loss of its residual star-forming gas is a long-standing problem of cluster dynamics (Hills 1980; Lada & Lada 2003; Boily & Kroupa 2003a; Baumgardt & Kroupa 2007). When the stars and the gas share the same density profile, the classical instantaneous-loss virial argument requires a global star formation efficiency of $\text{SFE} > 0.5$ for the remnant to stay bound, while N -body experiments, in which the cluster re-virializes and keeps a bound core, lower this to $\text{SFE} \approx 0.33$ (Boily & Kroupa 2003a,b; Baumgardt & Kroupa 2007). The star formation efficiencies inferred for embedded clusters are typically well below this value, so the identical-profile picture predicts near universal dissolution – in tension with the observed population of bound clusters.

The resolution, recognised well before the present framework, is that a stellar component *more centrally concentrated*

than the gas survives expulsion far more readily. Adams (2000) built semi-analytic models in which the stars follow a lowered-isothermal (King) distribution function in the combined star–gas potential – the construction we adopt here – with an ad hoc untruncated isothermal gas halo of free amplitude, and found substantial survival well below the classical threshold; N -body surveys with static gas potentials (Adams et al. 2006; Proszkow & Adams 2009) confirmed the picture. What these models left unspecified is the *origin* of the gas–star density contrast: the gas profile and its normalization were free parameters rather than consequences of the star formation process.

Parmentier & Pfalzner (2013) closed that gap. Star formation at a constant efficiency per free-fall time, ε_{ff} (Krumholz & McKee 2005), inside a centrally concentrated clump necessarily leaves a stellar component *steeper* than the residual gas: the local SFE is centrally peaked. In Shukirgaliyev et al. (2017, hereafter S17) we showed with direct N -body models that Plummer clusters formed this way sur-

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vive instantaneous gas expulsion down to $SFE \approx 0.15$, an effective SFE (Sect. 3.1) of ≈ 0.32 , insensitively to the Galactic tide (Shukirgaliyev et al. 2019). In Shukirgaliyev et al. (2021, hereafter S21) we extended this to Dehnen (1993) profiles, where survivability improves with a shallower outer slope and a steeper cusp: the lowest surviving efficiency sampled, $SFE_J = 0.01$ within the Jacobi radius, was reached by the $\gamma = 2$ and 2.5 models.

Fully coupled (magneto)hydrodynamical+ N -body simulations of cluster formation with star-by-star feedback have matured rapidly (e.g. Wall et al. 2019; Lahén et al. 2020; Grudić et al. 2021; Hirai et al. 2021; Fujii et al. 2021; Dinnbier & Walch 2020). They constrain two ingredients of the present framework: feedback-driven gas expulsion proceeds on roughly the free-fall time of the star-forming clump (Li et al. 2019), dynamically close to the instantaneous limit (Baumgardt & Kroupa 2007), and Fujii et al. (2021) find that star-by-star cluster formation with feedback yields cluster masses and radii matching a comparable calculation that assumed instantaneous star formation and gas expulsion (Fujii & Portegies Zwart 2016), concluding that the simplified treatment suffices for the cluster mass function and for the dynamical evolution after gas expulsion. What such simulations cannot yet deliver is statistics over initial-condition families. In our own models of this channel the wall-clock cost per realization grew from about a week at the coarsest refinement to seven months at the finest, so ensembles of ten realizations were affordable only at the coarsest level, and the outcome is highly sensitive to the stochastic details of massive-star feedback – when the first very massive star forms, its mass and position, and the adopted jet model – so that identical initial conditions yield markedly different cluster structures (Assilkhani et al. 2026b,a; Akhmetali et al. 2026). Statistics over realizations are therefore mandatory rather than optional, and mapping survivability across a family of initial structures requires tens of runs per parameter combination. The semi-analytic framework developed here is complementary to the coupled runs: they calibrate the sub-grid ingredients (ε_{ff} , the expulsion timescale), while it maps the parameter space with the statistics the problem demands.

The S17 and S21 initial conditions share two limitations. First, a Plummer stellar profile formed at constant ε_{ff} leaves a residual gas profile steeper than the dust-continuum profiles of observed young clumps. Second, and more basic, the Plummer and Dehnen profiles are spatially infinite, so the “global” SFE depends on the aperture within which it is measured – S17’s value was effectively aperture-limited, and S21 had to quote the SFE within the Jacobi radius and convert. Real embedded clusters, moreover, are routinely described by King models (King 1966), and the forthcoming N -body campaign (Paper I, in prep.) adopts them.

This paper therefore builds a King-family initial-condition set. Both the stars and the residual gas are finite – truncated together at the tidal radius – so the global SFE is a single unambiguous number, and the concentration W_0 becomes a free axis along which to map survivability. We work out, fully analytically up to quadratures, the structure and post-expulsion virial state of these models; check the virial machinery against the published S17/S21 thresholds; add a fully independent distribution-function estimate of the bound fraction; and derive the survivability predictions Paper I will test.

The paper is organized as follows. Section 2 constructs the King-family initial conditions and their local-SFE structure. Section 3 develops the post-expulsion virial state, reduces the sur-

vival condition to a single dimensionless parameter, compares the King predictions with the published families, and gives the threshold curve and its mechanism. Section 4 computes the bound fraction from the frozen distribution function and verifies it with Monte Carlo realizations. Section 5 introduces the free-fall-time truncation criterion. We discuss the implications in Sect. 6 and conclude in Sect. 7. The appendix documents the numerical verification of every quadrature and inversion used.

A note on terminology. Every SFE at which a cluster is deemed to survive in this paper is the output of a specific static criterion, not a dynamical result, and we name it accordingly. Two of these live in global-SFE space and are functions of W_0 : SFE_{vir} , the *virial threshold*, is the global SFE at which the post-expulsion virial ratio reaches an adopted survival value of the effective SFE, $eSFE_{\text{surv}}$; and SFE_{DF} , the *frozen-DF threshold*, is the global SFE at which the bound fraction of the frozen distribution function falls below $F_b = 0.02$. The survival value $eSFE_{\text{surv}}$ is a quantity of a different kind: a single adopted number in effective-SFE space (we take the classical 1/3 throughout), the *input* to the virial criterion, whereas SFE_{vir} is its *output* and runs from 0.225 to 0.106 across the $W_0 = 3\text{--}9$ grid of Paper I.

A convention for the SFE itself is needed because the families compared here are not all finite. For the King models both components are truncated at r_t , so the global SFE $SFE = M_*/(M_* + M_g)$ is unambiguous. The published Plummer and Dehnen models are spatially infinite, and we follow S17 and S21 in measuring their SFE within ten scale radii, SFE_{10} . Throughout the paper we write simply SFE, meaning the total value for King models and SFE_{10} for the published families; axes and tables are labelled accordingly and the distinction is flagged again only where the two are plotted together.

We reserve the term *critical SFE*, SFE_{crit} , for the dynamical threshold of a live cluster as measured in direct N -body simulations. It is not computed anywhere in this paper: SFE_{vir} and SFE_{DF} are static estimates of it, and the N -body campaign of Paper I is what will measure it. That measurement will itself be a prediction rather than a fact about real clusters, inheriting the idealizations of instantaneous expulsion and of gas that is removed rather than evolved; but it is the closest proxy available, and it is the number the estimates made here are to be judged against.

2. King-family initial conditions

2.1. Stellar King profile

The stellar cluster immediately before gas expulsion is a King model of concentration W_0 (King 1966). The construction of stars in virial equilibrium within the total star–gas potential follows Adams (2000); the essential difference is that here the gas profile is not free but is fixed by the star formation physics (Sect. 2.2), which ties the density contrast, the truncation, and the global SFE together. Our W_0 parametrizes the *stellar* King profile, whereas the well depth ψ_0 of Adams (2000) is defined in the total potential.

We present the structural figures in units $G = M_* = r_h = 1$, with M_* the total stellar mass and r_h the stellar half-mass radius. This matches the companion N -body campaign, which fixes the tidal filling factor $\lambda \equiv r_h/R_J$ across W_0 – i.e. it equates half-mass radii, not tidal radii – so a half-mass-radius normalisation is the one in which the models are actually compared. (S17 and S21 used core-radius and scale-length units respectively; the underlying King solution is computed in $G = M_* = r_0 = 1$ with r_0 the King scale radius, then rescaled.) Every SFE and $eSFE$

160 quantity below is dimensionless and invariant under the choice of length unit (Sect. 2.2). For reference, the models used in the campaign have $(W_0, c \equiv \log_{10} r_t/r_0, r_t/r_h) = (3, 0.672, 3.73)$, $(6, 1.255, 6.80)$, and $(9, 2.119, 8.54)$.

2.2. Residual gas

Star formation proceeds for a duration t_{SF} at a constant efficiency per free-fall time ε_{ff} (Krumholz & McKee 2005; Parmentier & Pflanzner 2013). We adopt $\varepsilon_{\text{ff}} = 0.01$ as a fiducial value motivated by measurements on molecular-cloud and larger scales (Krumholz et al. 2019; Utomo et al. 2018; Pokhrel et al. 2021) – reported values span roughly an order of magnitude, and Parmentier & Pflanzner (2013) themselves normalized at $\varepsilon_{\text{ff}} \simeq 0.1$ – and explore the sensitivity in Sect. 5, which also shows that every untruncated result below is exactly independent of this choice. As shown below, this choice affects only the free-fall-time truncation of Sect. 5; every other result is exactly independent of ε_{ff} . Defining

$$k \equiv \sqrt{\frac{8}{3\pi}} \varepsilon_{\text{ff}} t_{\text{SF}}, \quad (1)$$

the local Parmentier & Pflanzner (2013) relation between the stellar density ρ_* and the residual gas density ρ_g inverts in closed form. With $\alpha \equiv k^4 \rho_*^2$ and

$$\begin{aligned} K_0 &= [\alpha^3 + 36\alpha^2 + 216\alpha + 24\alpha \sqrt{3(\alpha + 27)}]^{1/3}, \\ K_1 &= \sqrt{\frac{\alpha^2 + \alpha(K_0 + 24) + K_0(K_0 + 12)}{12k^4 K_0}}, \\ K_2 &= \frac{(\alpha - K_0 + 24)(K_0 - \alpha)}{3k^4 K_0}, \end{aligned} \quad (2)$$

180 the gas density is

$$\rho_g = \frac{1}{k^2} - \frac{\rho_*}{2} - \frac{1}{2} \sqrt{K_2 + \frac{8}{k^6 K_1}} + K_1. \quad (3)$$

Equation (3) has two informative limits. Where the stellar density is high the gas saturates at a constant plateau, $\rho_g \rightarrow 1/k^2$, and the local SFE $\rho_*/(\rho_* + \rho_g) \rightarrow 1$: the cluster centre is gas-poor. Where the stellar density is low, $\rho_g \propto \rho_*^{2/3}$: the gas is shallower than the stars and dominates the outskirts. Because $\rho_g \rightarrow 0$ as $\rho_* \rightarrow 0$, the gas of a King cluster is truncated together with the stars at the King radius r_t ; the gas mass, and hence the global SFE $\text{SFE} = M_*/(M_* + M_g)$, are finite and fully set by (W_0, t_{SF}) .

A simultaneous rescaling $\rho \rightarrow c\rho$, $k \rightarrow c^{-1/2}k$ leaves Eq. (3) form-invariant and merely relabels t_{SF} ; every dimensionless result below is therefore independent of the unit convention. For the same reason ε_{ff} and t_{SF} enter only through k : every untruncated result of this paper is *exactly* independent of the adopted ε_{ff} (Sect. 5).

2.3. Structure of embedded King clusters

Figure 1 shows the stellar and gas density profiles of the $W_0 = 3, 6, 9$ models at a fixed global SFE = 0.17 (panel a) and the corresponding local SFE profiles for $W_0 = 0.5$ –20 (panel b). The behaviour anticipated in Sect. 2.2 is apparent: the local SFE approaches unity at the centre, falls through the global value near the half-mass radius, and the gas – everywhere shallower than

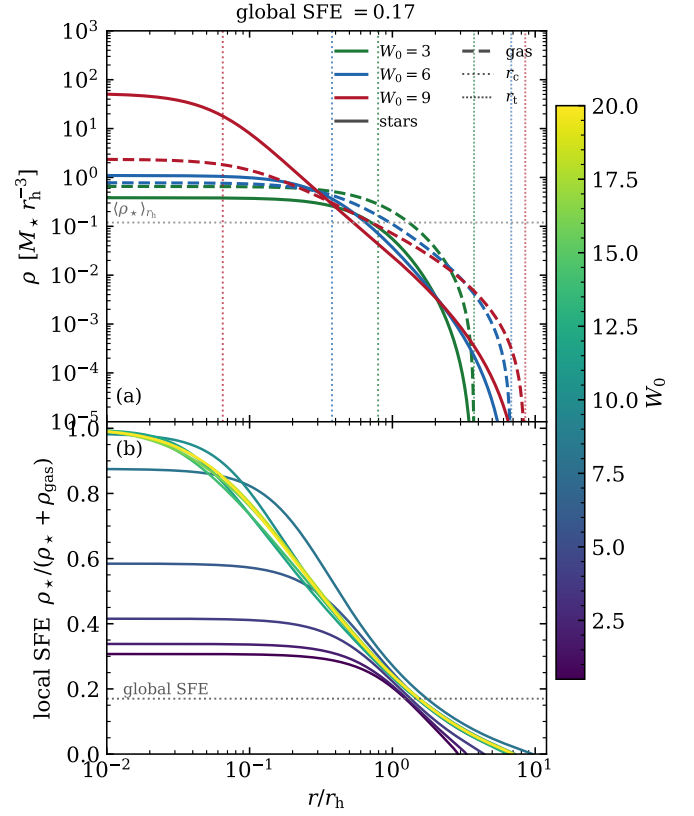


Fig. 1. King structure at a global SFE = 0.17, in units $G = M_* = r_h = 1$. (a) Stellar (solid) and residual gas (dashed) densities for $W_0 = 3, 6, 9$; coloured verticals mark r_0 (dotted) and r_t (fine dotted), the grey line the mean stellar density within r_h , $3/(8\pi)$. (b) Local SFE $\rho_*/(\rho_* + \rho_g)$ for $W_0 = 0.5$ –20; dotted line, the global value. See Sect. 2.3.

the stars – is truncated with them at r_t , which for these concentrations lies at $r_t/r_h \simeq 4$ –9. Both components being finite, the global SFE $\text{SFE} = M_*/(M_* + M_g)$ is a single well-defined number, not an aperture-dependent one. The mapping between t_{SF} and the global SFE for the whole family is given in Fig. 2.

3. Post-expulsion virial state and the SFE threshold

3.1. Effective SFE

The stars are taken to be in virial equilibrium within the total (stars + gas) potential at the instant of gas expulsion (S17). The scalar virial theorem for a steady-state stellar system in the total potential gives the stellar kinetic energy $2T = \int \rho_* \mathbf{r} \cdot \nabla(\Phi_* + \Phi_g) dV = |W_{\text{ss}}| + W_{\text{sg}}$, with

$$|W_{\text{ss}}| = 4\pi G \int_0^{r_t} \rho_*(r) M_*(r) r dr, \quad (4)$$

$$W_{\text{sg}} = 4\pi G \int_0^{r_t} \rho_*(r) M_g(r) r dr, \quad (5)$$

where $M_*(r)$ and $M_g(r)$ are the enclosed stellar and gas masses. The stellar density appears in both integrands because the virial theorem is applied to the stellar component: each integrand is the position-weighted gravitational force per unit volume acting on the stars, sourced by the enclosed stellar mass in Eq. (4) and by the enclosed gas mass in Eq. (5); the gas density enters only through $M_g(r)$. Equations (4)–(5) are exact for an arbitrary spherical gas distribution: the gas term follows from the shell

theorem alone, with no assumption that the gas traces the stars. (W_{sg} is the virial cross term, $\int \rho_{\star} \mathbf{r} \cdot \nabla \Phi_{\text{g}} dV$, which coincides with the mutual potential energy only in special cases; the virial theorem requires the former.) We verified this identity numerically for our strongly non-identical profiles: the kinetic energy obtained by integrating the frozen distribution function agrees with $(|W_{\text{ss}}| + W_{\text{sg}})/2$ to 1.6×10^{-3} for the $W_0 = 6$, SFE = 0.17 model, in which the gas cross term amounts to 190 per cent of the stellar self-energy term; this residual is set by the kinetic-energy quadrature rather than by the Eddington inversion, which is itself accurate to $\sim 10^{-5}$ (Appendix A). Immediately after expulsion the virial ratio is $Q = T/|W_{\text{ss}}|$, and the effective SFE (Goodwin 2009) is

$$e\text{SFE} \equiv \frac{1}{2Q} = \frac{|W_{\text{ss}}|}{|W_{\text{ss}}| + W_{\text{sg}}}. \quad (6)$$

If the gas traces the stars, $M_{\text{g}}(r) = [(1 - \text{SFE})/\text{SFE}] M_{\star}(r)$ and Eq. (6) recovers $e\text{SFE} = \text{SFE}$: the classical identical-profile case of Boily & Kroupa (2003a) is a *limit* of the present formulation, not an assumption of it. We use this limit as a numerical check of the implementation, satisfied to machine precision. The only ingredient inherited from identical-profile studies is the numerical reference value of the survival criterion, $e\text{SFE}_{\text{surv}}$, whose non-universality across density profiles is demonstrated in Sect. 3.3 and addressed dynamically in Sect. 4.

3.2. Reduction to a single parameter

Writing the cross energy as $W_{\text{sg}} = \eta (M_{\text{g}}/M_{\star}) |W_{\text{ss}}|$ defines the dimensionless gas *harmfulness*

$$\eta \equiv \frac{W_{\text{sg}} M_{\star}}{M_{\text{g}} |W_{\text{ss}}|}, \quad (7)$$

normalized so that gas tracing the stars gives $\eta = 1$. Then

$$e\text{SFE} = \frac{\text{SFE}}{\text{SFE} + \eta(1 - \text{SFE})}, \quad \text{SFE}_{\text{vir}} = \frac{\eta e\text{SFE}_{\text{surv}}}{1 - e\text{SFE}_{\text{surv}} + \eta e\text{SFE}_{\text{surv}}}$$

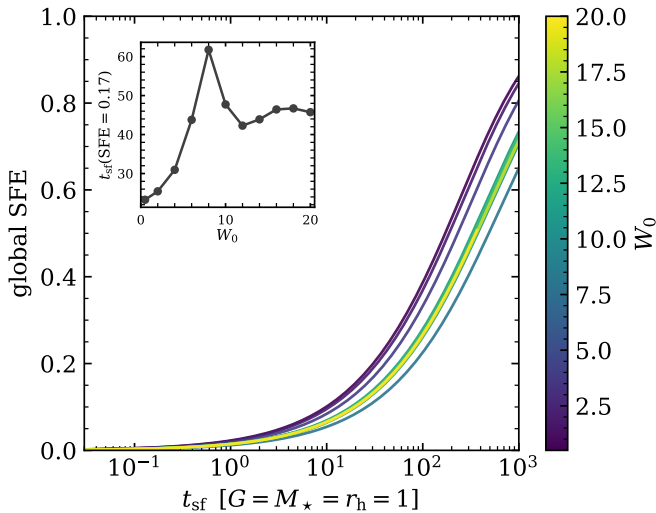


Fig. 2. Global SFE against the star formation duration for the King family, with time in units $G = M_{\star} = r_h = 1$ and the fiducial $\varepsilon_{\text{ff}} = 0.01$. The curves lie within a factor of ≈ 2.7 of one another across $W_0 = 0.5$ –20 (inset): at $\varepsilon_{\text{ff}} = 0.01$ reaching SFE = 0.17 takes $t_{\text{SF}} \approx 23$ ($W_0 = 0.5$) to ≈ 63 ($W_0 = 8$), i.e. ≈ 15 –40 free-fall times of the half-mass region ($t_{\text{ff}}(r_h) \approx 1.6$ in these units) – the low efficiency, not the geometry, sets the absolute scale.

Table 1. Effective SFE at the published survival thresholds.

Model	SFE _{thr}	$e\text{SFE}$ (this work)	$e\text{SFE}$ (published)
Plummer	0.150	0.317	0.32
Dehnen $\gamma = 0$	0.090	0.154	0.16
Dehnen $\gamma = 1^a$	0.024	0.060	0.06
Dehnen $\gamma = 2^a$	0.015	0.083	0.08

Notes. All four are infinite-extent families, so SFE_{thr} is SFE₁₀, the SFE within ten scale radii. ^(a) Lowest simulated SFE at which clusters survived (upper limit on the true threshold), published in S21 as SFE_J = 0.01, the SFE within the Jacobi radius. The SFE₁₀ values are converted here using the S21 setup ($\lambda \equiv r_h/R_J = 0.05$, hence $R_J = 20 r_h$, i.e. $R_J/a = 48.3$ and 20.0 for $\gamma = 1, 2$) and the gas model of Sect. 2.2; the resulting $e\text{SFE}$ values reproduce those quoted in S21.

for any adopted survival value $e\text{SFE}_{\text{surv}}$; for $e\text{SFE}_{\text{surv}} = 1/3$, $\text{SFE}_{\text{vir}} = \eta/(\eta + 2)$. All profile and concentration dependence of the survival threshold is carried by the single number η , evaluated at the threshold configuration. By the shell theorem only gas *enclosed* by a star contributes to Eq. (5); η therefore measures the radial segregation of the gas from the stars, which is the geometric quantity to follow in Sect. 3.4.

3.3. King and the published families

With the King machinery in hand we can place its predicted thresholds on the same axes as the published Plummer (S17) and Dehnen (S21) results. Table 1 first checks the virial machinery itself: the $e\text{SFE}$ computed here at each published N -body threshold reproduces the value quoted in those papers to a few per cent. Two conventions matter. Following S21, the global SFE of the infinite-extent profiles is measured within ten scale radii (SFE₁₀); the S17 Plummer threshold is reproduced ($e\text{SFE} = 0.317$ vs. 0.32) only under this aperture convention, confirming that the S17 SFE was effectively aperture-limited – a limitation the finite King models do not share. The $\gamma \geq 1$ Dehnen thresholds are the *lowest simulated* SFEs at which clusters still survived, i.e. upper limits (marked in Table 1).

Figure 3 places the King curves against the published families in the $e\text{SFE}$ –SFE plane. Every King model lies above the identical-profile diagonal $e\text{SFE} = \text{SFE}$: the centrally peaked SFE always makes gas expulsion less damaging than in the classical picture. Were $e\text{SFE}_{\text{surv}} = 1/3$ a universal survival threshold, every published marker would sit on it; the Plummer point nearly does, but the Dehnen thresholds fall well below, from $e\text{SFE} \approx 0.15$ ($\gamma = 0$) to $\lesssim 0.08$ for the cuspy models. *The virial survival value of $e\text{SFE}$ is therefore not universal across density profiles.* We nonetheless adopt the classical $e\text{SFE}_{\text{surv}} = 1/3$ for a first-pass King threshold curve (Sect. 3.4); Sect. 3.5 shows that no cumulative refinement of the virial statistic recovers universality, and the dynamical calculation of Sect. 4 is what closes the gap.

3.4. The King virial threshold curve and its mechanism

Adopting the classical $e\text{SFE}_{\text{surv}} = 1/3$, the King virial threshold is the solid “no truncation” curve of Fig. 9 (Sect. 5, where the free-fall-time truncation is added as the dashed curve). It is *non-monotonic*: it decreases from 0.249 at $W_0 = 0.5$ to a minimum of $\text{SFE}_{\text{vir}} = 0.106$ at $W_0 = 8.8$, rises to a local maximum of 0.137

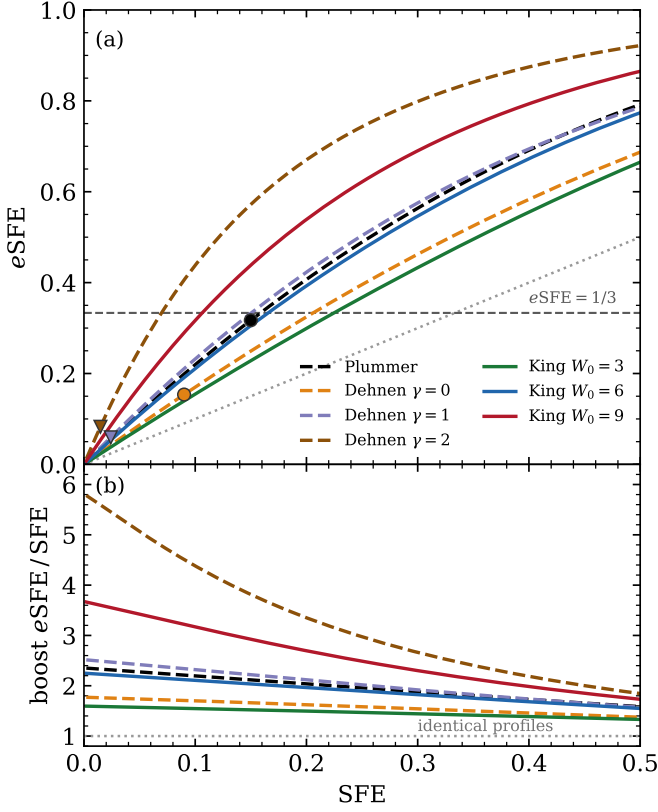


Fig. 3. (a) Effective against global SFE. Solid, the King models of this paper; dashed, the published families, with circles at their measured thresholds and triangles where the published value is a lowest-simulated upper limit. The horizontal line is $eSFE_{\text{surv}} = 1/3$; if it were universal every marker would lie on it. (b) The same curves as the boost factor $eSFE/SFE$, which resolves the low-SFE corner and tends to $1/\eta$ as $SFE \rightarrow 0$; dotted line, identical profiles. The abscissa is the total SFE for King and SFE_{10} for the published families throughout. See Sect. 3.3.

near $W_0 \simeq 13.5$, and settles around 0.13 at high W_0 . The simulated grid of Paper I ($W_0 = 3, 6, 9$; predicted thresholds 0.225, 0.166, 0.106; grey verticals in Fig. 9) brackets the descent into the minimum.

Figure 4 traces the causal chain behind the minimum. Panel (c) verifies the exact reduction of Eq. (8): the measured threshold curve coincides with $\eta/(\eta + 2)$ to numerical precision, so the question reduces to the behaviour of $\eta(W_0)$ (panel b), which reaches a minimum of $\eta = 0.238$ at $W_0 = 8.8$ – there the residual gas binds the stars roughly four times less effectively than identically distributed gas would. The same quantity is what Fig. 3b displays from the other side: the boost factor $eSFE/SFE$ tends to $1/\eta$ as $SFE \rightarrow 0$, so its intercept reads η off directly, giving $1/\eta = 1.59, 2.25$ and 3.66 for $W_0 = 3, 6$ and 9 . Panel (a) shows the geometric origin: the gas–star segregation, measured by $r_{h,\text{gas}}/r_{h,*}$, peaks (at a value of 3.0) near $W_0 \simeq 8.2$, closely tracking the well-known maximum of the King structural ratio r_t/r_h at $W_0 \simeq 8.0$.

The physical picture is as follows. With rising W_0 the King model develops a dense core, and the local-SFE relation places the residual gas in the increasingly extended, stellar-poor envelope: by the shell theorem, gas outside the bulk of the stars does not bind them, and η falls. Beyond $W_0 \simeq 8$, however, the stellar half-mass radius itself migrates rapidly outward into the developing r^{-2} halo – this is precisely why r_t/r_h turns over – so a growing fraction of stars sits in the gas-rich envelope, the seg-

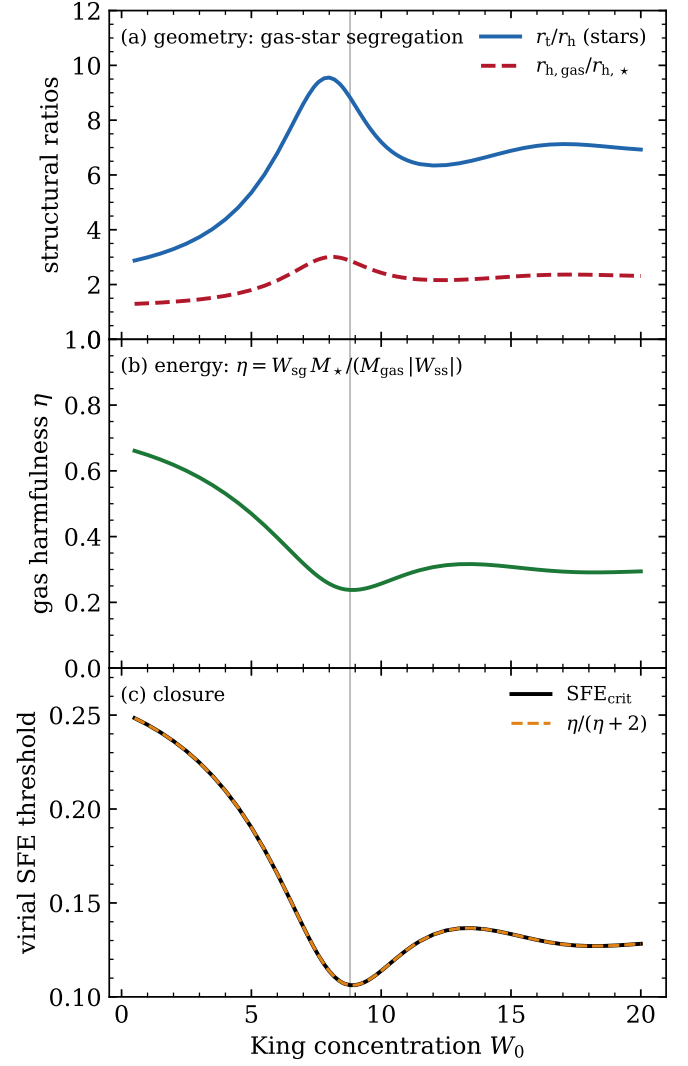


Fig. 4. Mechanism of the non-monotonic SFE threshold. (a) Structural ratios: r_t/r_h of the stars and the gas–star segregation $r_{h,\text{gas}}/r_{h,*}$ (scaled $\times 5$) at the threshold configuration. (b) Gas harmfulness η (Eq. 7). (c) Algebraic consistency check: the computed SFE_{vir} coincides with the exact reduction $\eta/(\eta + 2)$. The vertical line marks the minimum at $W_0 = 8.8$.

regation reverses, and η rises again, saturating as the model approaches self-similar isothermal structure. The small offsets between the extrema (segregation at $W_0 = 8.2$; η and SFE_{vir} at 8.8) arise because η weights the segregation by the energy integrand of Eq. (5) rather than by mass alone.

Because r_t/r_h is a property of the King family alone – independent of the star formation model – we expect a resilience optimum near $W_0 \simeq 8$ –9 in gas expulsion scenarios in which the residual gas remains preferentially extended relative to the stars, whatever mechanism produces that segregation. The argument does not by itself extend past our other idealizations – spherical symmetry, velocity isotropy, and instantaneous removal – whose relaxation is a question for the N -body campaign.

3.5. No universal cluster-wide criterion

A single cluster-wide $eSFE_{\text{surv}}$ is a coarse survival criterion: it is the dense, gas-poor *centre* of a concentrated model that stays bound, not the cluster as a whole. Figure 5 shows the cumulative

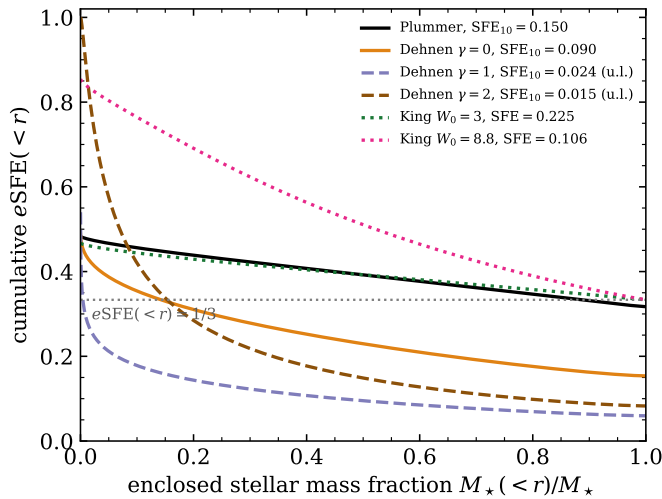


Fig. 5. Cumulative effective SFE, $eSFE(< r)$ – the ratio of virial integrals taken inside r , not an enclosed star formation efficiency – as a function of the enclosed stellar mass fraction, evaluated at each published family’s survival threshold (dashed: censored thresholds). Dotted curves show King models at their predicted thresholds ($eSFE_{\text{surv}} = 1/3$). The curves do not share a common crossing, so no cluster-wide static energy criterion is universal across profiles.

effective SFE $eSFE(< r)$ against the enclosed stellar mass fraction at each published family’s survival threshold. The curves do not share a common crossing, so no cluster-wide static energy statistic is universal across profiles.

A natural refinement is to ask whether a fixed *fraction* of the stellar mass – the part in a sufficiently bound inner region – is the profile-independent invariant. It is not. The stellar mass fraction interior to the radius where the enclosed SFE falls to $1/3$ is 0.54 for Plummer but only 0.08, 0.002 and 0.07 for Dehnen $\gamma = 0, 1, 2$ at their measured thresholds; measured instead against $eSFE(< r) = 1/3$ the fractions are 0.89, 0.14, 0.004 and 0.15. Either way the fraction scatters by more than an order of magnitude and tracks how far each global $eSFE$ sits from $1/3$ rather than revealing a new constant. The virial statistic, cluster-wide or as a cumulative inner-core fraction, therefore cannot by itself set the King threshold; a dynamical calculation is needed, which we provide in Sect. 4.

4. Bound fraction from the distribution function

The virial statistic of Sect. 3.1 characterizes the cluster as a whole and, as Sect. 3.3 showed, cannot by itself account for the survival of centrally concentrated models at low global $eSFE$. Following Adams (2000) and Boily & Kroupa (2003a) we therefore also compute the bound fraction directly from the stellar distribution function (DF). This is a second *static* estimator, not a dynamical calculation: the stellar positions are held fixed while the gas potential is removed and the escape condition is applied, so violent relaxation, revirialization and the recapture of marginally unbound stars are all absent by construction. Closing that gap requires the live N -body campaign of Paper I.

4.1. Method

The isotropic DF $f(\epsilon)$ is obtained by Eddington inversion (e.g. Binney & Tremaine 2008) of the stellar density in the total (star + gas) relative potential Ψ_{tot} . Rather than differencing a tabulated

$\rho_*(\Psi)$ twice – which is ill-conditioned for concentrated models, since the second-difference stencil is set by the inversion resolution and refining it amplifies noise – we evaluate the two required derivatives $d\rho_*/d\Psi$ and $d^2\rho_*/d\Psi^2$ in closed form via the chain rule through radius. The King density derivatives $\dot{\rho}'(W)$ and $\dot{\rho}''(W)$ are known analytically (and checked symbolically); dW/dr and d^2W/dr^2 are read directly from the King structure equation; and the potential derivatives follow from Gauss’s law, $\Psi'(r) = -M(r)/r^2$ and $\Psi''(r) = -4\pi\rho(r) + 2M(r)/r^3$. The remaining energy integral in Eddington’s formula is evaluated on a geometric grid after a $q = \epsilon - \Psi_t$ substitution that removes the integrable square-root singularity analytically. The inversion reproduces the analytic Plummer DF to a maximum relative error of 6×10^{-6} (Appendix A).

At expulsion the DF is frozen, and a star is bound if $v^2/2 < \Psi_b(r)$, where Ψ_b is generated by the bound stars only. Removing unbound stars shallows the potential, so the criterion is iterated to convergence at fixed stellar positions, with the density of retained stars capped by the initial stellar density. Every inversion is checked for self-consistency by reconstructing ρ_* from the recovered DF; an evaluation is accepted only if its reconstruction error stays below one per cent over $0.01 < M_*/M < 0.95$, and is otherwise recomputed on a refined energy grid (a $2000 \rightarrow 8000 \rightarrow 32000$ -point ladder, applied only where needed). With this scheme every continuum DF threshold used in the paper passes the one-per-cent check: the King family at $W_0 = 0.5$ – 20 , the four published families, and the dense $W_0 = 2.5$ – 20 scan behind Figs. 8 and 7, with a worst-case reconstruction error of 1.0 per cent.

This frozen-DF estimate neglects violent relaxation, revirialization, and the recapture of outflowing stars. For identical star and gas profiles it reproduces the known sharp analytic transition, retaining a bound remnant only for $SFE \gtrsim 0.4$ – 0.5 (Boily & Kroupa 2003a); it is thus *conservative* relative to N -body results in that regime.

4.2. King family: DF versus virial threshold

Figure 6 shows the resulting bound fraction $F_b(SFE)$ for the King family, and Fig. 8 compares the DF-based threshold (defined by $F_b < 0.02$) with the virial one. The level $F_b = 0.02$ is an operational definition of the frozen-DF transition, not a claim about a universal survival fraction. Where $F_b(SFE)$ rises steeply – $W_0 \lesssim 10$, covering the whole non-monotonic descent and the $W_0 = 3, 6, 9$ grid of Paper I – moving the cut between 0.01 and 0.05 shifts the threshold by at most 0.007 in SFE; for the most concentrated models the rise is gradual and the shift reaches 0.023, so those values are indicative only. The non-monotonicity survives at every cut in that range, with only the location of the broad minimum drifting between $W_0 \simeq 10$ and $\simeq 16$. The two criteria measure different physics and diverge strongly with concentration. For $W_0 \lesssim 5$ the DF threshold lies *above* the virial one – frozen-DF conservatism dominates – while for concentrated models it falls far below it, to $SFE_{\text{DF}} \simeq 0.048$ at $W_0 = 9$ and a broad minimum of $\simeq 0.018$ near $W_0 \simeq 13$, against virial values of 0.106 and 0.129. The two criteria therefore have their minima at different concentrations – $W_0 = 8.8$ for SFE_{vir} (Sect. 3.4) and $W_0 \simeq 13$ for SFE_{DF} – and the two should not be conflated: the first is where gas–star segregation is most effective, the second where the surviving central seed is densest. The dense, gas-poor centre retains a self-bound seed even when the cluster as a whole is formally super-virial. This is precisely the mechanism required by Sect. 3.3: applied to the published thresholds, the frozen-DF estimate leaves the cuspy Dehnen $\gamma = 2$ model

marginally bound ($F_b \approx 0.03$) at its measured threshold, while dissolving the shallower Plummer and Dehnen $\gamma = 0, 1$ cases – reproducing the *ordering* of the published thresholds across profiles, though not their exact values. The DF-based thresholds of the published families are $SFE_{DF} = 0.193$ (Plummer), 0.158 ($\gamma = 0$), 0.079 ($\gamma = 1$) and 0.009 ($\gamma = 2$), i.e. 1.3 and 1.8 times the measured Plummer and Dehnen $\gamma = 0$ N -body thresholds (Sect. 4.4).

4.3. Monte Carlo verification and finite- N effects

We checked the frozen-DF predictions against Monte Carlo realizations of the actual initial conditions: for each (W_0 , SFE) configuration, $N = 10^5$ equal-mass particles were sampled from the quasi-spherical DF in the total potential with AGAMA (Vasiliev 2019) – the same machinery that generates the N -body initial conditions of Paper I – and the iterated escape criterion applied particle-wise. Every quoted particle result uses 150 realizations at this N ; smaller N appears only in the scaling experiment below.

On the developed branch ($F_b \gtrsim 0.1$) the continuum agrees with the realizations to better than one per cent absolute at $W_0 = 3, 9, 12$ (Fig. 7), validating the inversion, the sampling and the criterion together. The 2–4 per cent excess at $W_0 = 6$ is not an inversion error: the continuum turns on $\Delta SFE \approx 0.012$ earlier in SFE than the discrete systems, and where the curve is near-vertical – as it is most sharply at $W_0 = 6$ – that horizontal offset reads as a vertical gap. The same offset is negligible at $W_0 = 9, 12$, and so is the gap.

The threshold is taken as the SFE at which the survival fraction $p = P(F_b > 0.02)$ crosses one half. Because the transition is narrow at $N = 10^5$ – at $W_0 = 3$ it completes within $\Delta SFE = 0.005$ – the SFE grid was refined around each crossing until several points fell inside the transition, so that the interpolation is not the limiting uncertainty. The error is then obtained by bootstrapping over realizations: the 150 outcomes at each SFE are resampled with replacement, p is re-formed and the crossing re-interpolated, and we quote the 16th–84th percentile spread. This propagates the binomial sampling noise of every grid point through the interpolation, and it does not assume a centre for the strongly bimodal cells, where a mean F_b has none. The result, marked in Fig. 8, is

$$SFE_{DF} = 0.3028, 0.1897, 0.0477, 0.0193 \quad (W_0 = 3, 6, 9, 12), \quad (9)$$

with bootstrap uncertainties of 1.1, 1.3, 0.6 and 0.6×10^{-4} respectively. An independent estimate, $\sigma_p/|dp/dSFE|$ with $\sigma_p = \sqrt{p(1-p)/150}$, agrees with the bootstrap to about 30 per cent. These are the values used wherever frozen-DF Monte Carlo thresholds are quoted (Sects. 6.6 and 7).

That precision belongs to the estimator, not to the prediction. The frozen-DF thresholds sit 1.3–1.8 times above the measured N -body values for the published families (Sect. 4.4), a model-level discrepancy some three orders of magnitude larger than these error bars.

The transition is intrinsically stochastic at finite N , and the stochasticity is itself N -dependent. Holding the configuration fixed just below the large- N threshold and varying only the particle number, the survival probability falls monotonically over more than two decades:

N	10^3	3×10^3	10^4	3×10^4	10^5	3×10^5
$W_0 = 3$	0.58	0.49	0.35	0.11	0.00	0.00
$W_0 = 9$	0.63	0.49	0.23	0.03	0.00	0.00

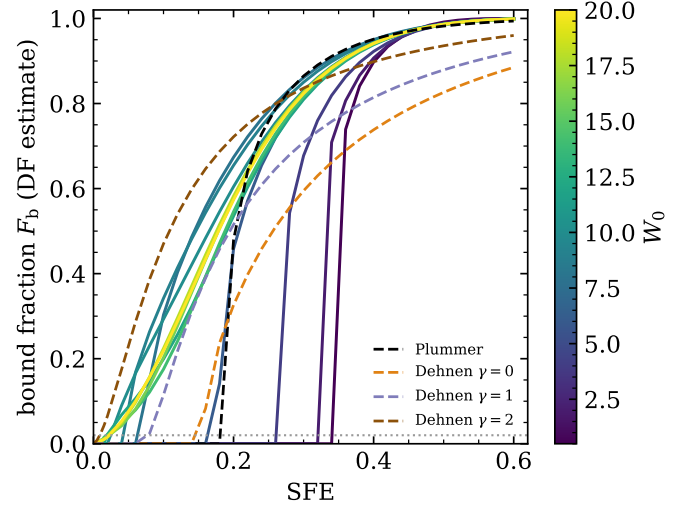


Fig. 6. Bound fraction after instantaneous gas expulsion, estimated from the frozen distribution function with the iterated escape criterion, as a function of the global SFE for King models with $W_0 = 0.5$ –20. The global SFE here is measured within the tidal radius (the infinite-extent published families elsewhere in this paper use SFE_{10}). The dotted line marks the $F_b = 0.02$ level used to define the DF-based threshold.

(150 realizations per entry, binomial errors ≤ 0.041 ; $W_0 = 3$ at $SFE = 0.30$ and $W_0 = 9$ at $SFE = 0.045$). Shot noise in the potential of the bound subset occasionally carries a sub-threshold realization into the basin of the surviving branch, and the chance of that happening vanishes as the sampling noise does. A diffuse and a concentrated model give the same decline, so this is a property of the family rather than of one configuration. The same stochasticity, augmented by violent relaxation, will govern the marginal cells of the N -body grid ($N \approx 1.7 \times 10^4$ in Paper I), for which the survival *probability* across realizations, rather than a sharp threshold, is the observable to measure.

The threshold itself, however, does not move with N . Repeating the measurement at $N = 3 \times 10^5$ shifts it by -8×10^{-5} at $W_0 = 3$ and $+7 \times 10^{-5}$ at $W_0 = 6$ – smaller than the bootstrap uncertainties quoted above, and of opposite sign, as scatter about a converged value rather than a drift. What tripling N does change is the *width* of the transition, which falls from $\Delta SFE = 0.0030$ to 0.0017 at $W_0 = 3$ and from 0.0020 to 0.0011 at $W_0 = 6$ (the SFE interval over which p runs from 0.1 to 0.9). Both ratios, 0.565 and 0.557, match the $N^{-1/2}$ expected of pure sampling noise ($3^{-1/2} = 0.577$) to within four per cent. Finite N therefore smears a fixed threshold symmetrically rather than displacing it, which is why the median crossing is the appropriate estimator: a definition keyed to the onset of survival would track the tail of that smearing and drift with N . This was verified at $W_0 = 3$ and 6, so the convergence of Eq. (9) is established at those two concentrations and assumed at the other two, where the transition has the same character. It is also the same effect as the table above seen from the other side – at a fixed SFE below the centre, a narrowing distribution leaves less probability above the survival cut, so p falls with N .

4.4. Confrontation with N -body results

For the Plummer and Dehnen $\gamma = 0$ families the bound fraction has been measured directly in N -body simulations, so the frozen-DF estimate can be tested rather than merely bracketed (Fig. 7). Two things follow.

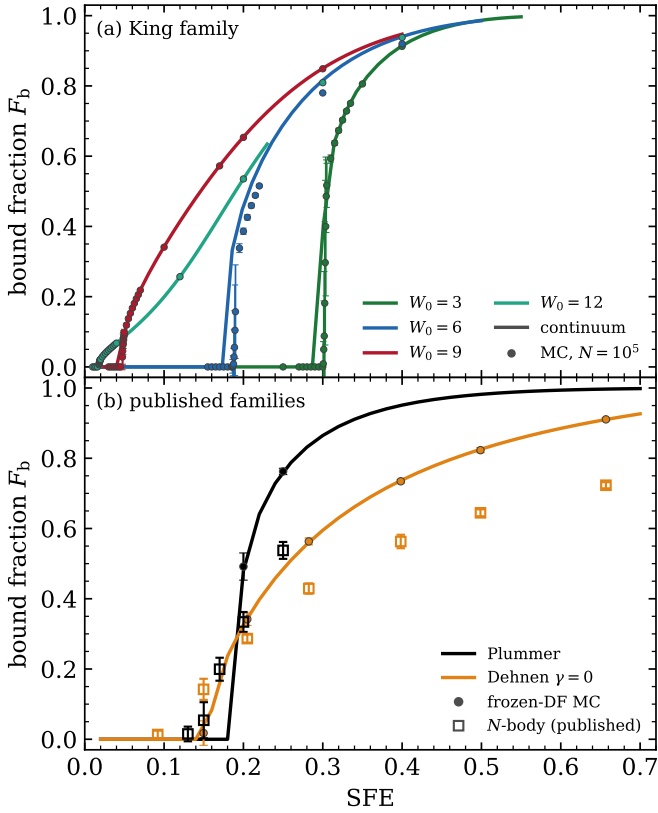


Fig. 7. Verification of the frozen-DF bound fraction. Curves, the continuum calculation; points, mean $\pm$ standard deviation over 150 realizations of $N = 10^5$ particles; the abscissa is the total SFE in (a) and SFE_{10} in (b). (a) King family (Sect. 4.3). (b) The two published families with N -body measurements, whose results are the open squares (Sect. 4.4).

First, the Monte Carlo and the continuum agree with each other to $\lesssim 0.01$ in F_b at every simulated SFE, and their published thresholds to about one per cent (0.193 vs. 0.192 for Plummer; 0.158 vs. 0.155 for Dehnen $\gamma = 0$). That rules out sampling noise and the marginal-branch sensitivity of Sect. 4.3 as explanations for the offset from the N -body points, leaving the physics the frozen distribution function omits as the natural candidate – though the comparison alone cannot exclude residual differences in the initial conditions or in how bound mass is measured.

Second, that separation is systematic and changes sign at the threshold. Below it the frozen DF dissolves clusters that survive in the simulations (at $SFE_{10} = 0.17$ the Plummer runs retain $F_b \approx 0.2$ where both frozen-DF estimates give zero); above it the frozen DF overpredicts the surviving mass, by +0.15 and +0.22 at $SFE_{10} = 0.20$ and 0.25 for Plummer, and by a nearly constant +0.13–+0.19 across $SFE_{10} = 0.28$ –0.66 for Dehnen $\gamma = 0$ (from a dedicated Monte Carlo comparison at the published N -body SFE values; Fig. 7). Both signs share one origin. The frozen DF holds the stars at fixed positions and lets the potential shallow as members are removed, so it misses the revirialization that lets a marginal cluster re-form a bound core – hence the missing survivors below threshold – while also ignoring the further mass loss revirialization imposes on a cluster that does survive – hence the excess above it. The corresponding thresholds are 1.3 and 1.8 times the measured N -body values for Plummer and Dehnen $\gamma = 0$ respectively.

Underlying both signs is a structural property of the frozen estimate: it is bistable, and the live system is not. The escape con-

dition is solved self-consistently – which stars are bound given the potential of the stars that are bound – and that iteration is self-amplifying, since unbinding a star shallows the potential and unbinds more. It therefore is observed to settle on one of only two outcomes, complete dissolution or a self-gravitating remnant, with no intermediate state persisting: a configuration retaining a small fraction of its stars keeps unbinding. Below a critical SFE the second outcome disappears, and F_b therefore does not rise continuously from zero. We report this empirically, as a property of the iteration we solve; the fixed-point analysis that would establish the bifurcation structure formally has not been carried out here.

The particle realizations show this directly. Through the entire transition at $W_0 = 3$ every realization either dissolves completely or retains $F_b \approx 0.45$ –0.52, with nothing in between: at $SFE = 0.3025$, 58 of 150 realizations survive with $F_b = 0.469 \pm 0.062$ and the other 92 give exactly zero. What sweeps across the transition is the *proportion* of survivors, not the bound fraction of a survivor, so the ensemble mean rises smoothly only as a mixture of zeros and halves and describes no individual cluster. The iteration count carries the same signature: convergence takes 16 iterations well below threshold and 11 well above it, but over 130 at the transition, the critical slowing down expected where two fixed points merge.

Live N -body clusters do not behave this way – the published Plummer sequence passes smoothly through $F_b = 0.015, 0.054, 0.199, 0.334$ and 0.538 as SFE_{10} runs from 0.13 to 0.25. Violent relaxation and revirialization let a remnant settle at an intermediate bound fraction that is dynamically reachable but is not a fixed point of the frozen map. The discontinuity is therefore a property of the approximation rather than of the physics, and it is the reason the frozen-DF threshold sits above the measured one: at $SFE_{10} = 0.17$ the frozen estimate must return zero, because the only alternative available to it is a fully-formed remnant.

The correction is thus not a single number: it depends on the profile and on which side of the threshold one sits, exactly as the non-universality of Sect. 3.3 would suggest. We therefore quote

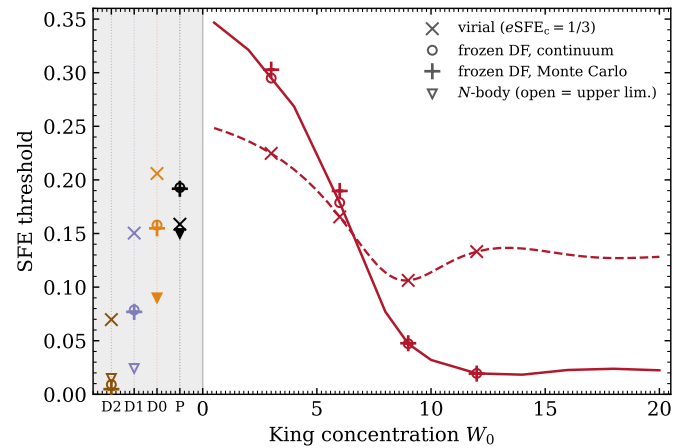


Fig. 8. SFE thresholds from every criterion considered. King family in red (dashed, virial; solid, frozen-DF continuum); the published families occupy the strip left of $W_0 = 0$ at ordinal positions (P, D0, D1, D2), where the abscissa carries no meaning. Symbols: $\times$ virial, $\circ$ frozen-DF continuum, $+$ frozen-DF Monte Carlo, ∇ published N -body (open where that value is an upper limit). The ordinate is criterion-specific; the shaded strip is where the SFE convention changes from the King total value to SFE_{10} . None of the plotted values is the critical SFE in the dynamical sense (Sect. 6.1).

the frozen-DF results as what they are – an upper bound on the surviving mass of a surviving cluster, and an empirical upper estimate of the critical SFE, calibrated on the published comparison rather than proved – and expect the N -body thresholds of Paper I to fall below the Monte Carlo values of Sect. 4.3, by a factor of order 1.3–1.8 if the published families are any guide.

5. A free-fall-time truncation criterion

The criterion of this section is a sensitivity experiment: we impose a hard boundary at $t_{\text{ff}} = t_{\text{SF}}$ to bound how much the results can depend on gas that had no time to form stars. It is not a claim that star formation switches off discontinuously at that density, and the untruncated model of Sect. 2.2 remains the baseline throughout.

The local-SFE relation of Sect. 2.2 presumes that the gas at every radius has had time to reach the local star formation efficiency per free-fall time. Where the local free-fall time exceeds the star formation duration, $t_{\text{ff}}(r) > t_{\text{SF}}$, this presumption fails. The condition $t_{\text{ff}} = t_{\text{SF}}$ translates into a scale-free contrast against the central gas plateau of Eq. (3), $\rho_{\text{plat}} \equiv k^{-2} = 3\pi/(8G\varepsilon_{\text{ff}}^2 t_{\text{SF}}^2)$, namely

$$\frac{\rho_{\text{SF}}}{\rho_{\text{plat}}} = \frac{\varepsilon_{\text{ff}}^2}{4}, \quad (10)$$

independent of t_{SF} , of the clump mass, and of all radial scales: the truncation is scale-free and controlled by ε_{ff} alone. At the observationally favoured $\varepsilon_{\text{ff}} = 0.01$ the contrast is only 2.5×10^{-5} , so the star-forming region extends to very low gas density and the truncation is a small correction; at the legacy $\varepsilon_{\text{ff}} = 0.05$ it is a factor 25 stronger.

This is the only place ε_{ff} can matter. It enters the gas profile solely through $k = \sqrt{8/(3\pi)}\varepsilon_{\text{ff}} t_{\text{SF}}$ (Eq. 1), so a change of ε_{ff} is *exactly* equivalent to a relabelling of t_{SF} , and every untruncated result of this paper – the e SFE–SFE relations, the threshold curves, the location of the minimum – is rigorously independent of it. Figure 9b shows the truncated curve for $\varepsilon_{\text{ff}} = 0.01$, 0.05 and 0.10: as $\varepsilon_{\text{ff}} \rightarrow 0$ the truncation becomes irrelevant and the curve converges to the untruncated result (minimum SFE_{vir} = 0.108 at $W_0 = 9$ for $\varepsilon_{\text{ff}} = 0.01$, against 0.106 at $W_0 = 8.8$ untruncated), while larger ε_{ff} removes more envelope gas, raising the thresholds and shifting the minimum to higher concentration (0.16 at $W_0 \approx 10$ for $\varepsilon_{\text{ff}} = 0.05$; 0.25 at $W_0 \approx 13$ for 0.10). The non-monotonicity survives for every value considered. For the published families the same sweep sets the minimum reachable SFE₁₀ under the removal flavour: for ($\varepsilon_{\text{ff}} = 0.01, 0.05, 0.10$) it is (0.026, 0.123, 0.228) for Plummer, (0.029, 0.136, 0.253) for Dehnen $\gamma = 0$, (0.033, 0.153, 0.280) for $\gamma = 1$ and (0.041, 0.185, 0.329) for $\gamma = 2$.

Two implementations must be distinguished. In the *removal* flavour, gas at $r > r_{\text{SF}}$ (where $\rho_{\text{g}} = \rho_{\text{SF}}$) is excluded from the potential entirely. Applied to the Dehnen models of S21 – whose total gas mass otherwise diverges as $r^{1/3}$, Eq. (10) providing a physical regularization – the reachable SFE₁₀ then has a minimum of 0.029, 0.033 and 0.041 for $\gamma = 0, 1, 2$. The Dehnen $\gamma = 0$ threshold (SFE₁₀ = 0.090) is comfortably above this floor and is therefore realizable under the removal flavour; only the two $\gamma \geq 1$ cases – both censored upper limits – lie below it. (At the legacy $\varepsilon_{\text{ff}} = 0.05$ the floors rise to 0.123–0.185, which forbids the three Dehnen configurations but still leaves the Plummer threshold, SFE₁₀ = 0.150, above its floor of 0.123; only at $\varepsilon_{\text{ff}} = 0.10$ is every published configuration excluded.) In the *aperture* flavour the dynamics retain the full gas distribution and

r_{SF} only bounds the region within which the clump and its SFE are measured; a third variant additionally reassigns the stellar mass beyond r_{SF} to gas (those stars are not yet meant to have formed), which lowers the global SFE to the stellar mass fraction within r_{SF} times SFE₁₀. Neither compresses the factor-of-ten spread of the published thresholds at $\varepsilon_{\text{ff}} = 0.01$, where r_{SF} is large and SFE($< r_{\text{SF}}$) $\rightarrow$ SFE₁₀.

For the King family the gas is finite in any case, and because r_{SF} falls close to r_t the three variants very nearly agree: essentially no stellar mass lies beyond r_{SF} at threshold ($M_{\star}(< r_{\text{SF}})/M_{\star} \geq 0.999$ for $W_0 = 3$ –12), so the third variant lowers the global SFE by ≤ 0.0002 , and removal differs from the aperture reading by no more than the 0.002 quoted below. They are not identical, but the distinctions that matter for the published families are immaterial here. Figure 9 compares the threshold curve with and without the removal-flavour truncation. At $\varepsilon_{\text{ff}} = 0.01$ the two curves are nearly indistinguishable: truncation raises the threshold by at most 0.002 and does not move the minimum, which stays at $W_0 \approx 9$. The star-forming radius reaches close to the tidal radius, $r_{\text{SF}}/r_t = 0.81$ –0.98 at threshold, so the mild internal tension of the assumed King profile extending past r_{SF} (Sect. 6.5) is confined to the outermost few per cent of the cluster. The non-monotonicity – the central result of this paper – is untouched.

6. Discussion

6.1. Two static criteria and the true threshold

Section 3.5 showed that no static energy statistic – cluster-wide or as an inner-core fraction – is universal across density pro-

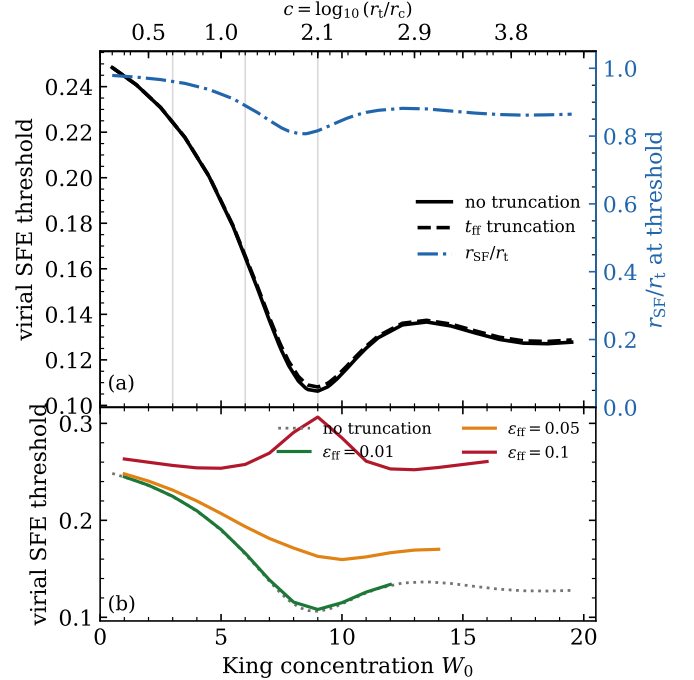


Fig. 9. (a) Virial SFE threshold with and without the free-fall-time truncation (removal flavour; black solid and dashed) at the fiducial $\varepsilon_{\text{ff}} = 0.01$, and the star-forming radius r_{SF}/r_t at the threshold configuration (blue dash-dot, right axis). Grey verticals mark the Paper I grid ($W_0 = 3, 6, 9$); the top axis carries $c = \log_{10}(r_t/r_c)$. (b) The truncated curve for $\varepsilon_{\text{ff}} = 0.01, 0.05$ and 0.10 against the untruncated result (dotted), which is exactly independent of ε_{ff} . See Sect. 5.

files, because survival of a concentrated system is decided by the dense, gas-poor centre that retains a self-bound seed even when the cluster as a whole is formally super-virial. The two criteria we can compute cheaply, the virial ratio and the frozen DF, therefore provide two complementary reference thresholds whose separation (Fig. 8) measures the physics that neither captures: violent relaxation, revirialization, and the recapture of out-flowing stars. Neither is a proven bound on the true threshold. What the published families do establish is one-sided: wherever the N -body threshold is measured rather than censored it lies below the frozen-DF value (Plummer 0.150 versus 0.193; Dehnen $\gamma = 0$, 0.090 versus 0.158), as does the $\gamma = 1$ upper limit; only the $\gamma = 2$ limit is uninformative. We therefore expect the King N -body thresholds to fall below their frozen-DF values as well. Figure 8 collects every criterion – virial, frozen-DF continuum, frozen-DF Monte Carlo, and the published N -body results – for the King family and the published families on a common axis. The two criteria cross near $W_0 \approx 5$ –6. For diffuse clusters the frozen-DF estimate is conservative (it disrupts the Plummer model at its measured N -body threshold) and the virial criterion is the closer guide; for concentrated clusters the situation reverses, with the virial criterion overestimating the threshold by about a factor of two at $W_0 = 9$ (0.106 versus 0.047). Paper I will locate the true N -body threshold relative to this pair; combining its grid with the S17/S21 points should then allow $eSFE_{\text{surv}}$ to be calibrated as an explicit function of a concentration measure, restoring the usefulness of the energy criterion in a profile-dependent form.

Finally, the Monte Carlo experiments of Sect. 4.3 show that close to threshold the very notion of a sharp critical SFE dissolves: survival becomes a probability over realizations, with shot noise able to rescue sub-threshold clusters at low N . This stochasticity is not a numerical artefact but a property of the discrete problem. We caution that the trend was measured at two concentrations only, and that the survival probability is a property of the frozen escape criterion rather than of a live cluster. Whether real low- N clusters, which span $N \sim 10^2$ – 10^6 , show a correspondingly shifted survival probability is a question for live N -body experiments rather than a result of this work.

6.2. Robustness to the expulsion timescale, ε_{ff} , and isotropy

Three model assumptions deserve scrutiny. First, instantaneous expulsion: coupled simulations find feedback-driven expulsion to proceed on roughly a clump free-fall time (Li et al. 2019), dynamically close to the impulsive limit (Baumgardt & Kroupa 2007), and Fujii et al. (2021) conclude, by comparing star-by-star cluster formation against the instantaneous star formation and gas expulsion of Fujii & Portegies Zwart (2016), that the simplified treatment is sufficient for the dynamical evolution of clusters after gas expulsion. Slower expulsion raises all bound fractions (Baumgardt & Kroupa 2007), so our thresholds are conservative in that respect; whether its W_0 -dependent magnitude preserves the non-monotonic *shape* is not established here, since a correction of one sign can still vary with concentration enough to move or erase an extremum. The predictions of this paper therefore apply specifically to the instantaneous-expulsion limit. Second, ε_{ff} : by the exact degeneracy of Sect. 5, every untruncated result is rigorously independent of it, and the truncated threshold curves remain non-monotonic across the full observationally reported range $\varepsilon_{\text{ff}} = 0.01$ – 0.10 (Fig. 9b). Third, isotropy: the initial conditions are isotropic by construction, matching the N -body models of Paper I; primordial velocity anisotropy would alter the frozen-DF bound fractions – Adam

(2000) showed radial anisotropy raises them – and is a natural extension of the framework that we defer.

6.3. Simulation support for the centrally peaked SFE

Beyond the analytic argument of Parmentier & Pfallner (2013), our MHD+ N -body simulations of this formation channel show, in the specific setups modelled there, that centrally concentrated clouds assemble centrally concentrated clusters – in fact *more* compact ones than the analytic construction predicts, with fitted Plummer and EFF scale radii systematically below the input value and a shape well described by a Plummer profile (Assilkhan et al. 2026b). This outcome arises through global collapse of the gas toward the centre rather than the purely local conversion assumed by Parmentier & Pfallner (2013), so the framework of Sect. 2.2 is supported at the level of the emergent structure even where its formation mechanism is idealized. The stellar substructure inherited from the cloud is erased within ~ 2.5 free-fall times by dynamical relaxation (Akhmetali et al. 2026), consistent with treating the cluster as a smooth centrally concentrated system by the time of gas expulsion. The concentration of the emergent cluster is, however, sensitive to early feedback physics: including protostellar jets in otherwise identical initial conditions yields systematically *less* concentrated outcomes – more extended, more substructured, less tightly bound, and more supervirial stellar systems, with the star formation efficiency suppressed from 19–33 to 12–16 per cent (Assilkhan et al. 2026a). Different feedback channels thus map one and the same clump onto a *range* of emergent concentrations – precisely the axis the King W_0 family parametrizes, and the reason cluster survivability must be known across the whole family rather than at a single fiducial profile.

6.4. Observational touchpoints

Three observational touchpoints already exist. First, the constant- ε_{ff} assumption inherited from Parmentier & Pfallner (2013) is directly supported within nearby clouds: Pokhrel et al. (2021) find $\Sigma_{\text{SFR}} = \varepsilon_{\text{ff}} \Sigma_{\text{gas}}/t_{\text{ff}}$ with ε_{ff} nearly independent of gas density, a median $\varepsilon_{\text{ff}} \approx 0.026$, and a cloud-to-cloud scatter of only 0.18 dex, consistent with the earlier surface-density relation of Gutermuth et al. (2011); on both sub-galactic and galaxy-wide scales the value converges to $\varepsilon_{\text{ff}} \approx 0.01$ (Krumholz et al. 2019; Utomo et al. 2018), which we adopt as fiducial. At the highest column densities the picture is debated, the CAFFEINE survey finding the SFE to saturate above a critical density (Mattern et al. 2024); but the exact ε_{ff} – t_{SF} degeneracy of the ε_{ff} – t_{SF} degeneracy renders every untruncated result of this paper immune to the controversy, and only the free-fall-time truncation of Sect. 5 is exposed to it. Second, the truncation itself offers a physical reading of the empirical dense-gas threshold for star formation (Lada et al. 2010, 2012): in our framework the apparent threshold is not a universal density but a free-fall-time condition, $\rho_{\text{SF}} = 3\pi/(32 G t_{\text{SF}}^2)$, predicting that the threshold should scale with the star formation duration of the clump. Third, the model predicts the density profile of the *primordial clump* itself: because the Parmentier & Pfallner (2013) construction holds the total density fixed while gas is converted into stars – an assumption built into Eq. (3) – the sum $\rho_{\star} + \rho_{\text{g}}$ is the initial clump profile, directly comparable to dust-continuum observations of clumps at early evolutionary stages. Fitting $\rho \propto r^{-p}$ between the radii enclosing 10 and 90 per cent of the mass yields $p_{\text{tot}} \approx 2.2$ – 2.5 across the entire King family, insensitive

to both W_0 and the global SFE. This matches the clump-scale profiles of the youngest observed sources ($p \approx 2.2$ at the earliest phases; Gieser et al. 2023), classical single-power-law fits giving a shallower $\langle p \rangle = 1.6\text{--}1.7$ with a documented outward steepening (Mueller et al. 2002; Beuther et al. 2002; Pirogov 2009). The near-universality of the slope is itself a prediction: the envelope slope cannot discriminate the concentration of the embedded cluster, which is instead encoded in the density contrast between the central plateau and the clump mean. A forward-modelled comparison with the ATLASGAL and Hi-GAL clump populations (Urquhart et al. 2018; Elia et al. 2021) is deferred to future work.

6.5. Towards a self-consistent stellar profile

An internal tension of the model remains. The stellar profile is *assumed* to be a King model extending to r_t , while the free-fall-time criterion of Sect. 5 implies that star formation is confined within $r_{\text{SF}} < r_t$ (Fig. 9). A self-consistent treatment would instead start from the primordial clump profile, apply the local-SFE relation together with the free-fall-time criterion, and *derive* the stellar profile – generally close to, but not exactly, a truncated King model – iterating the pair to a fixed point. We regard this as the principal structural refinement for future work. A second, more practical lesson of the same kind emerged in Sect. 3.3: quoted SFE values depend materially on the measurement aperture (the S17 threshold is reproduced only under the 10 a aperture, and the S21 thresholds required an explicit $\text{SFE}_J \rightarrow \text{SFE}_{10}$ conversion), so comparisons of star formation efficiencies across studies – observational or numerical – must state the aperture to be meaningful.

6.6. Predictions for the N -body campaign

The framework makes the following concrete, falsifiable predictions for the simulation grid of Paper I ($W_0 = 3, 6, 9$; $N \approx 1.7 \times 10^4$). (1) The virial criterion with $e\text{SFE}_{\text{surv}} = 1/3$ places the virial threshold at 0.225, 0.166, and 0.106 for $W_0 = 3, 6, 9$ (minimum 0.106 at $W_0 = 8.8$); at the fiducial $\varepsilon_{\text{ff}} = 0.01$ the free-fall-time truncation leaves these values unchanged to within 0.002. (2) The frozen-DF Monte Carlo thresholds are 0.3028, 0.1897 and 0.0477 (plus 0.0193 at $W_0 = 12$), each determined to $\sim 10^{-4}$; we expect the measured N -body thresholds to lie below the frozen-DF estimates, as they do for Plummer and Dehnen $\gamma = 0$; where they fall relative to the virial estimate is a prediction to be tested, not a bracket established here. In particular the *ordering* of survivability with W_0 – the non-monotonicity – is the sharpest qualitative test. (3) On the developed branch the bound-fraction curves of Fig. 7 are quantitative initial-condition-level predictions, verified against $N = 10^5$ realizations to better than one per cent for $W_0 = 3, 9, 12$ and a few per cent for $W_0 = 6$; systematic deviations of the N -body results from these curves isolate the contribution of violent relaxation. (4) In the marginal cells, survival should be reported as a probability over random realizations rather than a binary outcome; at $N \approx 1.7 \times 10^4$ the scaling above puts the expected survival probability near 0.2–0.3 at $W_0 = 3$, $\text{SFE} = 0.30$. (5) As a consistency check of the initial-condition pipeline, in the half-mass-radius units the campaign actually uses ($G = M_\star = r_h = 1$, with $\lambda \equiv r_h/R_J$ held fixed across W_0) the star formation duration required for a given SFE is nearly universal, $t_{\text{SF}}(\text{SFE}=0.17) \approx 23\text{--}63$ across the whole family (Fig. 2; factor ≈ 2.7), the absolute scale set by the low ε_{ff} rather than by the geometry.

7. Conclusions

We have built a King-family initial-condition set for embedded clusters formed with a centrally peaked star formation efficiency – both components finite and truncated together at the tidal radius, so the global SFE is a single unambiguous number – and used it to map two static estimators across W_0 : the virial threshold SFE_{vir} , whose calibration reproduces the effective SFEs inferred at the published N -body thresholds, and the independent frozen-DF threshold SFE_{DF} . Neither is the dynamical critical SFE, which only the N -body campaign can measure. Our results are as follows.

1. The SFE predicted by the virial criterion for surviving instantaneous gas expulsion is a non-monotonic function of the King concentration, with a minimum of $\text{SFE}_{\text{vir}} \approx 0.11$ at $W_0 \approx 8.8$ for the adopted $e\text{SFE}_{\text{surv}} = 1/3$. The ordering with W_0 , not this absolute value, is the falsifiable prediction.
2. The virial survival criterion reduces exactly to $\text{SFE}_{\text{vir}} = \eta e\text{SFE}_{\text{surv}} / (1 - e\text{SFE}_{\text{surv}} + \eta e\text{SFE}_{\text{surv}})$, where the gas harmfulness η (Eq. 7) carries the entire profile dependence. Within this model family the minimum of η lies close to the maximum of the King structural ratio r_t/r_h at $W_0 \approx 8$, where the gas–star segregation is greatest; the two extrema do not coincide exactly, because η weights the segregation by the cross-energy integrand rather than by mass.
3. The survival value of the effective SFE is not universal across density profiles, decreasing from $\approx 1/3$ (Plummer) to $\lesssim 0.08$ (cuspy Dehnen models); static energy criteria, global or cumulative, do not collapse onto a universal threshold.
4. Gas with a local free-fall time exceeding the star formation duration lies beyond a scale-free density contrast $\rho_{\text{SF}}/\rho_{\text{plat}} = \varepsilon_{\text{ff}}^2/4$. At the observationally favoured $\varepsilon_{\text{ff}} = 0.01$ this truncation leaves the King threshold curve unchanged to within 0.002; only at the legacy $\varepsilon_{\text{ff}} = 0.05$, a factor 25 stronger, does it forbid every published configuration, which is why ε_{ff} and the removal-versus-aperture choice matter only there.
5. A frozen-DF bound-fraction estimate in the spirit of Adams (2000), built on an analytic Eddington inversion with a per-evaluation reconstruction check and cross-checked against Monte Carlo realizations, reproduces the ordering of the published thresholds and predicts that concentrated King models keep a self-bound central seed down to global SFEs of a few per cent ($\text{SFE}_{\text{DF}} = 0.3028, 0.1897, 0.0477, 0.0193$ for $W_0 = 3, 6, 9, 12$). Near threshold the outcome of individual realizations is stochastic, with a sub-threshold rescue probability that falls with N . The two criteria are complementary rather than bounding, but the published families place the measured N -body threshold below the frozen-DF value, and we expect the same for King.
6. The non-monotonic prediction is robust to the truncation choice, and the $W_0 = 3, 6, 9$ grid of Paper I brackets the descent into the minimum, providing a direct test.

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all figures are publicly available at <https://github.com/BS-astronomer/kingmodels-public> and archived at doi:10.5281/zenodo.22646889.

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Appendix A: Numerical verification

King density near the truncation edge. The direct evaluation of the King density, $\hat{\rho}(W) = e^W \text{erf}(\sqrt{W}) - \sqrt{4W/\pi}(1 + 2W/3)$, suffers catastrophic cancellation near the edge: both terms scale as $\sqrt{W}$ while their difference is $O(W^{5/2})$, so for $W \lesssim 10^{-8}$ the direct expression returns pure round-off. We therefore use the series $\hat{\rho}(W) = \frac{8}{15\sqrt{\pi}} W^{5/2}(1 + \frac{2}{7}W + \frac{4}{63}W^2 + \dots)$ for $W < 10^{-3}$, where its truncation error is below 10^{-6} ; the first and second derivatives $\hat{\rho}'(W)$, $\hat{\rho}''(W)$ needed for the Eddington inversion are given the matching series and direct branches, each cross-checked against symbolic differentiation to $< 10^{-9}$. A closed-form large- α expansion is likewise substituted in the gas inversion (Eq. 3) for $\alpha = k^4 \rho_\star^2 > 10^8$, where the exact expression loses precision in the difference $K_0 - \alpha$.

Identical-profile and virial checks. The implementation passes the identical-profile limit ($\rho_g \propto \rho_\star \Rightarrow e\text{SFE} = \text{SFE}$) to machine precision. For non-identical profiles, the stellar kinetic energy integrated over the distribution function agrees with $(|W_{\text{ss}}| + W_{\text{sg}})/2$ to 1.6×10^{-3} at $W_0 = 6$, $\text{SFE} = 0.17$, where the gas cross term is 190 per cent of the stellar self-energy. The virial thresholds are converged in radial resolution: at the minimum the threshold moves by $< 10^{-7}$ between 1000 and 8000 grid points.

Eddington inversion and the bound-fraction estimate. Finite differencing a tabulated $\rho_\star(\Psi)$ twice is ill-conditioned for concentrated models, because the second stencil's width is set by the inversion resolution, so refining it amplifies noise. We instead obtain both derivatives in closed form by the chain rule through radius – $\hat{\rho}'$, $\hat{\rho}''$ analytically, dW/dr and d^2W/dr^2 from the structure equation, and $\Psi' = -M/r^2$, $\Psi'' = -4\pi\rho + 2M/r^3$ from Gauss's law – leaving one energy integral, evaluated on a geometric grid after the substitution $q = \epsilon - \Psi_t$ that removes the square-root singularity. The result reproduces the analytic Plummer distribution function to 6×10^{-6} , two orders of magnitude better than the chained finite-difference inversion. Each inversion is then checked by reconstructing $\rho_\star$ over $0.01 < M_\star/M < 0.95$, with failures recomputed on a refined grid (2000 $\rightarrow$ 8000 $\rightarrow$ 32000 points, applied only where needed). Every continuum threshold in this paper passes the one-per-cent check, worst case 1.0 per cent.

Positivity of the recovered distribution function. A reconstruction test can be passed while $f(\epsilon)$ itself turns negative, so positivity was checked separately across the sixteen King models ($W_0 = 0.5\text{--}20$, each at its own threshold) and the four published families. The published families are non-negative everywhere; every King model is non-negative at each *interior* energy, the sole exception being the final grid node $\epsilon = \Psi_{\text{tot}}(0)$, where the boundary term $\propto q^{-1/2}$ is evaluated – one node in 2000, in all sixteen cases. That node is harmless by construction: every phase-space integral weights f by $\sqrt{\Psi_{\text{tot}} - \epsilon}$, which vanishes identically there, so its contribution is exactly zero and the clip applied in production changes no reported quantity.